

IMPLICIT DIFFERENTIATION

Section 2.5

9-10-25

ex:

$$1) y = \pm \sqrt{x}$$

$$y' = \pm \frac{1}{2\sqrt{x}} = \boxed{\frac{1}{2y}}$$

$$2) x = (f(x))^2$$

$$1 = 2(f(x))' \cdot f'(x)$$

$$f'(x) = \frac{1}{2(f(x))} = \boxed{\frac{1}{2x}} = y'$$

We now use $\frac{dy}{dx}$, not $f'(x)$ or y'
like a slidy gear (yujina)

$$3) x = y^2$$

$$1 = 2(y)' \cdot \frac{dy}{dx}$$

$$\boxed{\frac{dy}{dx} = \frac{1}{2y}}$$

Tidbit of the day

$\frac{d}{dx} y$ ← derivative of:

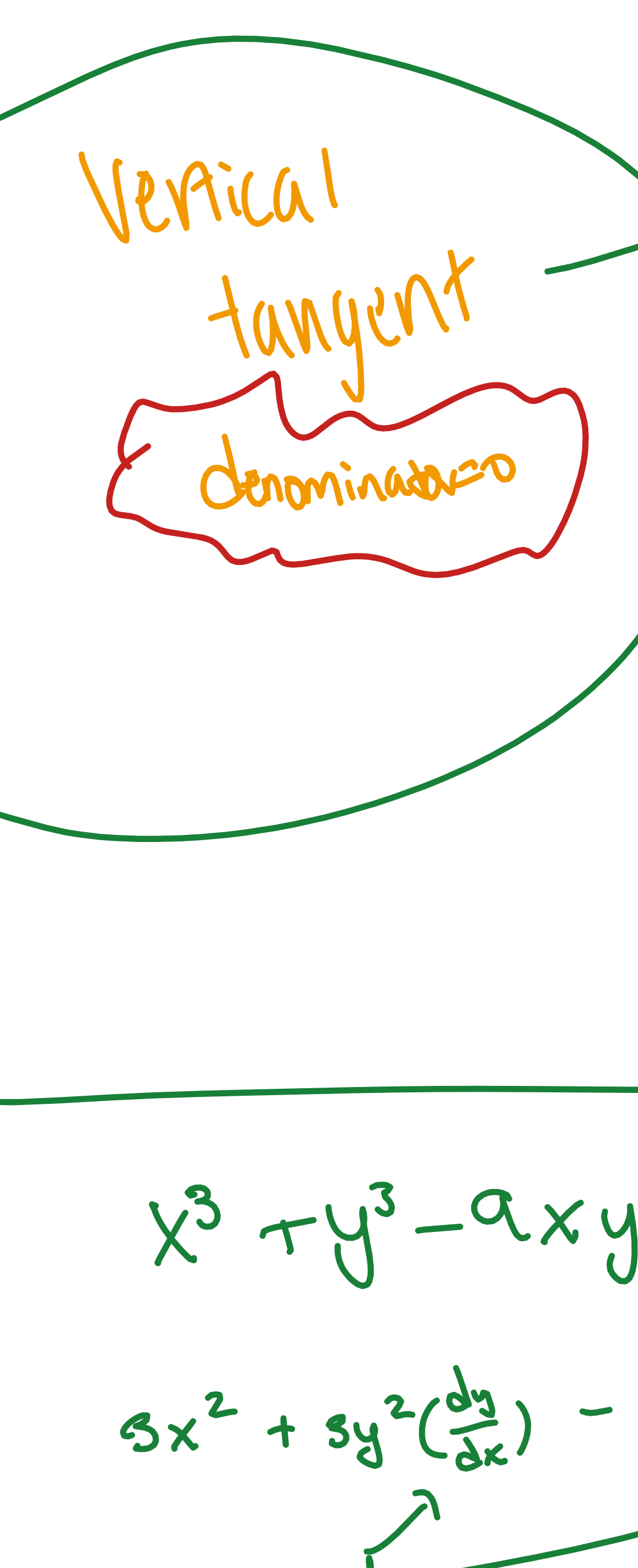
$\frac{d}{dx} x$ ← with respect to:

ex: $x^2 + y^2 = 4$

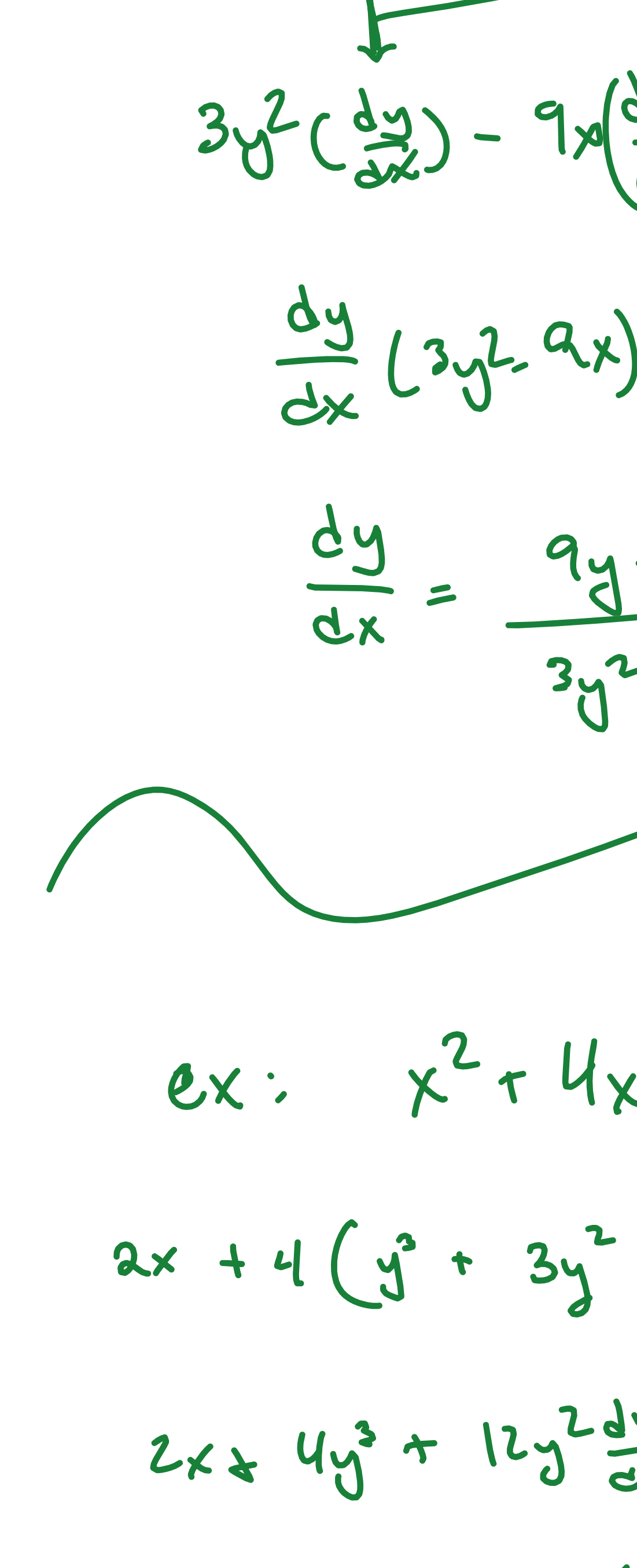
find $\frac{dy}{dx}$

$$2x + 2y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \boxed{-\frac{x}{y}}$$



@ $x=0$
↓
plug fs in
 $y^2 = 4$
 $y = \pm 2$
 $(0, -2)$
 $(0, 2)$



@ $y=0$
↓
plug fs in
 $x^2 = 4$
 $x = \pm 2$
 $(2, 0)$
 $(-2, 0)$

$$x^3 + y^3 - 9xy = 0$$

$$3x^2 + 3y^2 \left(\frac{dy}{dx}\right) - 9y - 9x \frac{dy}{dx} = 0$$

$$3y^2 \left(\frac{dy}{dx}\right) - 9x \left(\frac{dy}{dx}\right) = 9y - 3x^2$$

$$\frac{dy}{dx} (3y^2 - 9x) = 9y - 3x^2$$

$$\frac{dy}{dx} = \frac{9y - 3x^2}{3y^2 - 9x} = \boxed{\frac{3y - x^2}{y^2 - 3x}}$$

ex: $x^2 + 4xy^3 + 3y^2 = 8$

$$2x + 4(y^3 + 3y^2 \left(\frac{dy}{dx}\right)x) + 3(2y \cdot \frac{dy}{dx}) = 0$$

$$2x + 4y^3 + 12y^2 \frac{dy}{dx} x + 6y \frac{dy}{dx} = 0$$

$$2x + 4y^3 + 12xy^2 \left(\frac{dy}{dx}\right) + 6y \left(\frac{dy}{dx}\right) = 0$$

$$12xy^2 \left(\frac{dy}{dx}\right) + 6y \left(\frac{dy}{dx}\right) = -4y^3 - 2x$$

$$\left(\frac{dy}{dx}\right) (12xy^2 + 6y) = -4y^3 - 2x$$

$$\frac{dy}{dx} = \frac{-4y^3 - 2x}{12xy^2 + 6y}$$

$$= \frac{-2y^3 + x}{6xy^2 + y}$$

$\cos(y^2) + 3x^2 = 10$

$$\left(-\sin(y^2) \cdot 2y \cdot \frac{dy}{dx}\right) + 6x = 0$$

$$-2y \sin(y^2) \left(\frac{dy}{dx}\right) = -6x$$

$$\frac{dy}{dx} = \frac{-6x}{-2y \sin(y^2)}$$

$$\frac{dy}{dx} = \frac{3x}{y \sin(y^2)}$$

TIDBIT:

second deriv:

$$\frac{d^2y}{dx^2}$$

$\frac{d^2y}{dx^2}$ of $y' = 4x$

$$2y \left(\frac{dy}{dx}\right) = 4$$

$$\frac{dy}{dx} = \frac{4}{2y} = \frac{2}{y} = 2y^{-1}$$

$$\frac{d^2y}{dx^2} = -2(y)^{-2} \cdot \left(\frac{dy}{dx}\right) = \frac{-2}{y^2} \left(\frac{dy}{dx}\right)$$

That's not enough

$$-\frac{2}{y^2} \cdot \frac{2}{y} = \frac{-4}{y^3}$$

OR (keep going)

Boom! work

$$\frac{-4}{y \cdot y^2} = \frac{-4}{y \cdot 4x} = \boxed{\frac{-1}{xy}}$$

$2x^3 - 3y^2 = 8$ find $\frac{d^2y}{dx^2}$

$$6x^2 - 6y \left(\frac{dy}{dx}\right) = 0$$

$$\frac{dy}{dx} = \frac{-6x^2}{-6y} = \frac{x^2}{y}$$

$$\frac{d^2y}{dx^2} = \frac{2x \cdot y - \frac{dy}{dx} \cdot x^2}{y^2}$$